Server Backup Methods

Objectives:

At the end of this episode, I will be able to:

1. Identify server backup methods.

Exam Objective: *3.7 - Explain the importance of backups and restores.*

Description: In this episode, the viewer will identify server backup methods such as full, synthetic, incremental and differential backup types, as well as file-level vs. system-state backups.

- Data Backups
  - Most common types start with full backups
  - Content
    - User data
    - Application data
    - State data
  - Data creation and modifications
    - Since last backup
    - Turns on archive bit
    - Included in the next backup
- Common Backup Types
  - Full
    - All data is copied
    - Archive bit is not inspected
    - All archive bits are cleared
    - Pro's
      - Lowest backup set or sets
      - Fastest recovery time
    - Con's
      - Slowest backup type
      - Greatest storage accrual
  - Differential
    - Only data created or modified since the last full backup
    - Archive bit is inspected
    - Archive bit is not cleared
    - Pro's
      - Less time then full backup
      - Less storage accrual than full backup
      - Great for data sets that are to great to run a full backup
    - Con's
      - Greater backup sets than full backup
      - Slower than incremental
      - Larger storage cost than incremental
  - Incremental
    - Only data modified since the last full or incremental backup
    - Archive bit is inspected

        - Archive bit is cleared
        - Pro's
            - Fastest backup type
            - Lowest storage accrual
        - Con's
            - Longest restore time
            - Greatest backup sets

- Other "backup" methods
    - Open file
        - Hot snapshots
        - Windows VSS and previous versions
        - Running VM snapshot
    - Snapshot/Checkpoint
        - VM machine snapshots/checkpoints
    - Archive

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Not applicable if blank